(MATH101, 102, 311, 312) Calculus and Analysis

Problems

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Contents

1 (MATH101) Calculus I

Problems within each course are arranged chronologically by year and semester, then by institution in the order POSTECH, KAIST, and SNU, while preserving the original problem order within each source.

Problem 1.1. (2021 Spring KAIST Calculus I) Does each of the following infinite series converge? Mark Yes or No. You do not need to justify your answers.

(1) $\displaystyle\sum_{n=1}^{\infty} \frac{n!}{n^n}$

(2) $\displaystyle\sum_{n=1}^{\infty} \frac{e^{\sqrt{n}}}{n\sqrt{n}}$

(3) $\displaystyle\sum_{n=1}^{\infty} \left(1-\frac{1}{n}\right)^{n}$

(4) $\displaystyle\sum_{n=1}^{\infty} (-1)^n \left(1-\frac{1}{n}\right)^{n\ln n}$

(5) $\displaystyle\sum_{n=1}^{\infty} \left(1-\frac{1}{n}\right)^{2n\ln n}$

(6) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{(n+1)\ln(n+1)}$

(7) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{(n+1)\ln(n+1)\ln\ln(n+1)}$

(8) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{(n+1)(\ln(n+1))^2}$

(9) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{\tan\left(\arccos\left(\frac{1}{n+1}\right)\right)}$

(10) $\displaystyle\sum_{n=1}^{\infty} \frac{\sin(1/n)}{n^{\alpha}}, \qquad \alpha > 0$

Problem 1.2. (2021 Spring KAIST Calculus I) Find the area of the portion of the ellipse

$$9x^2 + 16y^2 = 144$$

between the y-axis and $x = 2$.

Problem 1.3. (2021 Spring KAIST Calculus I) Find the antiderivative of

$$\frac{1}{(x^2+x+1)(x^2+x+2)(x^2+x+3)}.$$

Problem 1.4. (2021 Spring KAIST Calculus I) Does the following improper integral converge?

$$\int_0^{\infty} \frac{\sin^3 x}{x}\, dx.$$

Problem 1.5. (2021 Spring KAIST Calculus I) Prove or disprove:

(1) If $\displaystyle\sum_{n=1}^{\infty} a_n$ converges, then $\displaystyle\sum_{n=1}^{\infty} a_n^2$ also converges.

(2) If both $\displaystyle\sum_{n=1}^{\infty} a_n$ and $\displaystyle\sum_{n=1}^{\infty} |b_n|$ converge, then $\displaystyle\sum_{n=1}^{\infty} a_n b_n$ also converges.

(3) Assume $a_n \neq 0$ for all n. If $\displaystyle\lim_{n\to\infty} a_n = 0$ and $\displaystyle\sum_{n=1}^{\infty} |\sin(a_n)|$ converges, then $\displaystyle\sum_{n=1}^{\infty} |a_n|$ converges.

Problem 1.6. (2021 Spring KAIST Calculus I) Evaluate

$$\int_1^\infty \left(2\sqrt{x^2-1} - 2x + \frac{1}{x}\right) \mathrm{d}x.$$

Problem 1.7. (2021 Spring KAIST Calculus I) Let f be a continuously differentiable (i.e., $f(x)$ is differentiable and $f'(x)$ is continuous) one-to-one function defined on $[a, b]$. Assume that $f'(x) \neq 0$ on $[a, b]$. Compute the following sum of two integrals:

$$\int_{f(a)}^{f(b)} f^{-1}(y)\ \mathrm{d}y + \int_a^b f(x)\ \mathrm{d}x.$$

Problem 1.8. (2021 Spring KAIST Calculus I) Let n be a positive integer and denote the following improper integral by a_n:

$$a_n = \int_0^1 (\ln x)^n\ \mathrm{d}x.$$

(a) Show that the improper integral a_n is convergent. Compute the value of a_n.

(b) Is the series

$$\sum_{n=1}^\infty \frac{1}{a_n}$$

absolutely convergent, conditionally convergent, or divergent? Explain the reason using appropriate tests.

(c) Is the series

$$\sum_{n=2}^\infty (-1)^n \frac{1}{\ln |a_n|}$$

absolutely convergent, conditionally convergent, or divergent? Explain the reason using appropriate tests.

Problem 1.9. (2021 Spring KAIST Calculus I) Show that the power series

$$\sum_{n=2}^\infty n(n-1)x^n, \qquad |x| < 1,$$

can be expressed as

$$\frac{x^2}{(1-x)^3}.$$

Problem 1.10. (2021 Spring KAIST Calculus I) Consider the standard ellipse C,

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \qquad a > b > 0.$$

Denote by e the eccentricity of C.

(a) Find an upper bound of the function

$$f(t) = \sqrt{1 - e^2 \sin^2 t}$$

using the first two terms of its binomial series expansion.

(b) Show that the perimeter (total arc length) of C is less than

$$2\pi a \left(1 - \frac{e^2}{4}\right).$$

Problem 1.11. (2021 Spring KAIST Calculus I) Consider the power series

$$\sum_{n=1}^{\infty} (-1)^n \frac{(x-1)^{2n+1}}{n2^n}.$$

(a) Find its interval of convergence.

(b) At each point in the interval of convergence, discuss its mode of convergence as absolute or conditional convergence together with proper reasoning.

Problem 1.12. (2021 Fall KAIST Calculus I) Is each of the following statements true or false? Mark T or F. You do not need to justify your answers.

(1) $f = O(g)$ implies $f = o(g)$ for functions f and g that are positive for x sufficiently large.

(2) $x(\ln x)^{2022} = o(x^{1.001})$.

(3) If $f'(a) \neq 0$, then there exists a $\delta > 0$ such that $f(x) \neq f(a)$ for all $x \neq a$ with $a - \delta < x < a + \delta$.

(4) $\displaystyle\int_2^{\infty} \frac{1 - 2^{-x}}{1 + \sqrt{x}}\, dx$ converges.

(5) $\displaystyle\int_1^{6} \frac{1}{x - 4}\, dx$ converges.

(6) $\displaystyle\int_2^{\infty} \frac{1}{\ln x}\, dx$ converges.

(7) $\displaystyle\int_1^{2} \frac{1}{x \ln x}\, dx$ converges.

(8) $\displaystyle\int_{-\infty}^{\infty} \frac{1}{e^x + e^{-x}}\, dx$ converges.

(9) $\displaystyle\int_{-\infty}^{\infty} \frac{2x}{x^2 + 1}\, dx$ converges.

(10) $\displaystyle\lim_{x \to \infty} \frac{\ln(x + 10^{2022})}{\ln x} = 1.$

Problem 1.13. (2021 Fall KAIST Calculus I) Evaluate

$$\int_1^{\infty} \left(2\sqrt{x^2 - 1} - 2x + \frac{1}{x}\right) dx.$$

Problem 1.14. (2021 Fall KAIST Calculus I) Let f be a continuously differentiable (i.e., $f(x)$ is differentiable and $f'(x)$ is continuous) one-to-one function defined on $[a, b]$. Assume that $f'(x) \neq 0$ on $[a, b]$. Compute the following sum of two integrals:

$$\int_{f(a)}^{f(b)} f^{-1}(y)\ \mathrm{d}y + \int_a^b f(x)\ \mathrm{d}x.$$

Problem 1.15. (2021 Fall KAIST Calculus I)

(a) Derive $\dfrac{\mathrm{d}}{\mathrm{d}x}(\operatorname{arctanh} x)$ for $|x| < 1$.

(b) Find $\displaystyle\int x \operatorname{arctanh} x\ \mathrm{d}x$.

Problem 1.16. (2021 Fall KAIST Calculus I) Let

$$\Gamma(n) = \int_0^\infty x^{n-1}e^{-x}\ \mathrm{d}x$$

for a positive integer n.

(a) Use a comparison test to show that $\Gamma(n)$ converges.

(b) Show that $\Gamma(n) = (n-1)!$.

Problem 1.17. (2021 Fall KAIST Calculus I) Let $w(t)$ be a differentiable function and let κ and ρ be positive constants.

(a) Solve

$$\frac{\mathrm{d}}{\mathrm{d}t}w(t) + \kappa w(t) - \rho = 0 \qquad \text{or} \qquad w' + \kappa w - \rho = 0$$

with the initial condition $w(0) = v_0$.

(b) By setting $w(t) = \dfrac{\mathrm{d}}{\mathrm{d}t}y(t)$, solve

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}y(t) + \kappa\frac{\mathrm{d}}{\mathrm{d}t}y(t) - \rho = 0 \qquad \text{or} \qquad y'' + \kappa y' - \rho = 0$$

with $y(0) = 0$ and

$$\left.\frac{\mathrm{d}}{\mathrm{d}t}y(t)\right|_{t=0} = v_0.$$

Problem 1.18. (2021 Fall KAIST Calculus I)

(1) Solve the equation

$$xy' + 2y = \frac{8x}{(x-1)(x+1)(x+2)}$$

for $x > 1$.

(2) Solve the initial value problem

$$y' = 2x\sqrt{1+y^2}, \qquad y(0) = 0.$$

Problem 1.19. (2021 Fall KAIST Calculus I) Determine the radius and interval of convergence of the following series.

(1) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n} x^n$

(2) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n!} x^n$

(3) $\displaystyle\sum_{n=0}^{\infty} (-1)^n x^{2^n}$

Problem 1.20. (2021 Fall KAIST Calculus I) Consider the power series

$$\sum_{n=1}^{\infty} (-1)^n \frac{(x-1)^{2n+1}}{n2^n}.$$

(a) Find its interval of convergence.

(b) At each point in the interval of convergence, discuss its mode of convergence as absolute or conditional convergence together with proper reasoning.

2 (MATH102) Calculus II

Problem 2.1. (2021 Spring KAIST Calculus I) Mark True or False for each of the following statements. You do not need to justify your answers.

(1) If a series $\sum_{n=0}^{\infty} a_n x^n$ converges on $[-1, 1]$, then the series $\sum_{n=1}^{\infty} n a_n x^{n-1}$ converges on $(-1, 1)$.

(2) If $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are three-dimensional vectors, then

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \times \mathbf{b}) \times \mathbf{c}.$$

(3) If $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are nonzero three-dimensional vectors and $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = 0$, then $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are in the same plane.

(4) Assume f is infinitely differentiable at 0 (i.e., $f^{(n)}(0)$ exists for all $n = 1, 2, 3, \ldots$). Then there exists $\varepsilon > 0$ such that

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

for all $-\varepsilon < x < \varepsilon$.

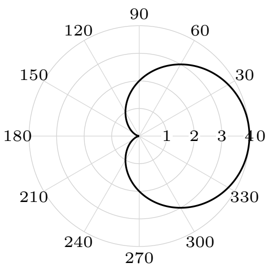
(5) Let $A(x) = \sum_n a_n x^n$ and $B(x) = \sum_n b_n x^n$ be power series with radii of convergence R_1 and R_2, respectively. Then $A(x)B(x)$ is again a power series with radius of convergence $\min\{R_1, R_2\}$.

Problem 2.2. (2021 Spring KAIST Calculus I) Match each of the following polar equations with an appropriate graph.

(1) $r = 4\cos(2\theta)$

(2) $r = \dfrac{4}{2 - \cos\theta}$

(3) $r = 2(1 + \cos\theta)$



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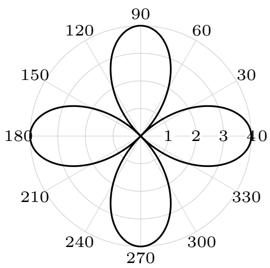

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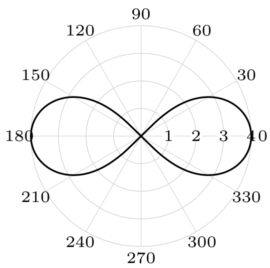

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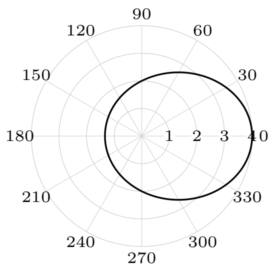


(4)

Problem 2.3. (2021 Spring KAIST Calculus I) Let

$$L_1 : \ x = 1 + t, \ y = 3 - t, \ z = 2t, \qquad L_2 : \ x = 0, \ y = -2s, \ z = 5s.$$

Find the minimum of $\|\overrightarrow{P_1 P_2}\|$ for $P_1 \in L_1$ and $P_2 \in L_2$.

Problem 2.4. (2021 Spring KAIST Calculus I) Let M_1 be the plane in space given by

$$2x - y - 3z = 2,$$

and let M_2 be the plane perpendicular to M_1 through the points $P(1, -2, -4)$ and $Q(0, -1, -2)$.

(a) Find parametric equations for the line L of intersection of M_1 and M_2.

(b) Find an equation for a plane M parallel to L that contains the line through the points $A(-2, -4, 3)$ and $B(2, 0, 4)$. Find the distance between L and M.

Problem 2.5. (2021 Spring KAIST Calculus I) For the surface with equation

$$(y - \sqrt{2z_0})^2 - x^2 = z,$$

consider its curve cut C by the plane $z = z_0 > 0$.

(a) Identify the type of C and find its eccentricity.

(b) Find the foci and directrices of the curve cut on the plane $z = z_0$.

(c) Find the polar equation $r = f(\theta)$ of the curve cut on the plane $z = z_0$.

Problem 2.6. (2021 Fall POSTECH Calculus II) Find an equation of the plane that passes through the line of intersection of the planes

$$x + y = 2 \qquad \text{and} \qquad 2x - y + 3z = 1$$

and passes through the point $(1, 2, 1)$.

Problem 2.7. (2021 Fall POSTECH Calculus II) Let $f(x, y) = |xy|$. Show that f is differentiable at $(0, 0)$.

Problem 2.8. (2021 Fall POSTECH Calculus II) Let $F(x, y, z)$ be differentiable everywhere. Assume that

$$F(1 + t^2, 2t, -1 - t) = 0$$

for all t, and that $\nabla F(2, 2, -3) = (a, b, c)$. Show that

$$a + b = \frac{c}{2}.$$

Problem 2.9. (2021 Fall POSTECH Calculus II) A plane P contains the points $(-3, 0, 0)$, $(0, -3, 0)$, and $(0, 0, -3)$. Find all points on

$$z = x^2 + y^2$$

whose tangent plane does not intersect P.

Problem 2.10. (2021 Fall POSTECH Calculus II) Find the absolute maximum and minimum values of

$$f(x, y) = x^4 + y^4 - 4xy + 1$$

on the region

$$x^4 + y^4 \leq 1.$$

Problem 2.11. **(2021 Fall POSTECH Calculus II)** Let L_1 and L_2 be lines in $\mathbb{R}^3$ parametrized by

$$L_1(t) = (1+t, 1-t, 2t), \qquad L_2(s) = (-s, s, -s),$$

respectively. Find the shortest distance between L_1 and L_2.

Problem 2.12. **(2021 Fall POSTECH Calculus II)** Let $f(x, y) = |xy|$. Show that there is no open ball centered at $(0, 0)$ in which f_x and f_y are defined and continuous.

Problem 2.13. **(2021 Fall POSTECH Calculus II)** Let

$$f(x, y) = \begin{cases} \dfrac{(xy)^\alpha}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

For which $\alpha > 0$ is f differentiable in $\mathbb{R}^2$?

Problem 2.14. **(2021 Fall POSTECH Calculus II)** Let

$$R = \left\{ (x, y) : 0 \leq x \leq \frac{\sqrt{3}}{2},\ 0 \leq y \leq \frac{\sqrt{3}}{2},\ x^2 + y^2 \leq 1 \right\}.$$

Find the minimum and maximum of the linear function

$$F(x, y) = -x + 2y.$$

Justify your answer.

Problem 2.15. **(2021 Fall POSTECH Calculus II)** Show that

$$f(x, y) = x^2 - 2xy^2 - y^4$$

has a saddle point at $(0, 0)$.

Problem 2.16. **(2021 Fall POSTECH Calculus II)** Show by giving a counterexample that the cross product is not associative; namely, find three vectors $\mathbf{U}, \mathbf{V}, \mathbf{W} \in \mathbb{R}^3$ such that

$$(\mathbf{U} \times \mathbf{V}) \times \mathbf{W} \neq \mathbf{U} \times (\mathbf{V} \times \mathbf{W}).$$

Problem 2.17. **(2021 Fall POSTECH Calculus II)** Determine the maximum and minimum values of

$$f(x, y) = xy \ln(1 + x^2 + y^2)$$

in the region

$$x^2 + y^2 \leq 1.$$

Problem 2.18. **(2021 Fall POSTECH Calculus II)** Evaluate

$$I_x = \iiint_D x \, \mathrm{d}V,$$

where

$$D = \left\{ (x, y, z) : 1 \leq x^2 + y^2 + z^2 \leq 4,\ x \geq 0 \right\}.$$

Problem 2.19. (2021 Fall POSTECH Calculus II) Evaluate the surface integral

$$\iint_S \mathbf{F} \cdot \mathbf{N} \, \mathrm{d}S$$

representing the flux of the vector field

$$\mathbf{F}(x, y, z) = (1 + x, x + 2y, -3x + z)$$

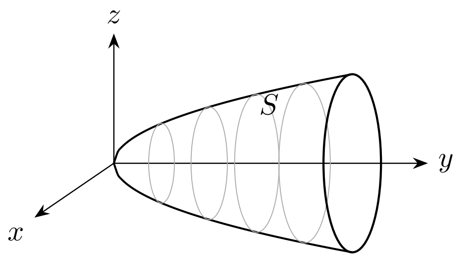
across the hemisphere

$$S = \left\{(x, y, z) : x^2 + y^2 + z^2 = 1, \ x \geq 0\right\}$$

in the direction of the unit normal vector field whose first component is positive.

Problem 2.20. (2021 Fall POSTECH Calculus II) Let $S \subseteq \mathbb{R}^3$ be the surface defined by

$$y = x^2 + z^2, \qquad x^2 + z^2 \leq 1.$$



Let

$$\mathbf{F}(x, y, z) = (y, z, x).$$

Find the inward flux of $\nabla \times \mathbf{F}$ across S in the direction of the unit normal vector field whose second component is positive.

Problem 2.21. (2021 Fall POSTECH Calculus II) Let

$$f(x, y) = |(x - 1)(y - 1)|.$$

Show that there is no open ball centered at $(1, 1)$ in which f_x and f_y are defined and continuous.

Problem 2.22. (2021 Fall POSTECH Calculus II) Find the volume of the solid E bounded by both the sphere

$$x^2 + y^2 + z^2 = 2z$$

and the sphere

$$x^2 + y^2 + z^2 = \frac{1}{4}.$$

Problem 2.23. (2021 Fall POSTECH Calculus II) An object moves from the origin to a point that lies on the surface

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

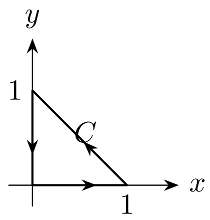
along a smooth curve C under the force field

$$\mathbf{F}(x, y, z) = (yz, zx, xy).$$

Find the maximum possible value of the work integral

$$\int_C \mathbf{F} \cdot \mathrm{d}\mathbf{r}.$$

Problem 2.24. (2021 Fall POSTECH Calculus II) Let C be the triangle shown below, oriented counterclockwise.



Evaluate

$$\oint_C x^2 \, dy.$$

Problem 2.25. (2021 Fall POSTECH Calculus II) Let $S \subseteq \mathbb{R}^3$ be a surface with a parametrization $\boldsymbol{\alpha}\colon \mathcal{R} \to S$ defined by

$$\boldsymbol{\alpha}(u, v) = uv\mathbf{i} + \frac{u}{v}\mathbf{j} + \frac{v}{u}\mathbf{k}.$$

Find an equation for the tangent plane to S at $\boldsymbol{\alpha}(1, 1)$.

Problem 2.26. (2021 Fall KAIST Calculus I) Mark True or False for each of the following statements. You do not need to justify your answers.

(1) If a series $\sum_{n=0}^{\infty} a_n x^n$ converges on $[-1, 1]$, then the series $\sum_{n=1}^{\infty} n a_n x^{n-1}$ converges on $(-1, 1)$.

(2) If $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are three-dimensional vectors, then

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \times \mathbf{b}) \times \mathbf{c}.$$

(3) If $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are nonzero three-dimensional vectors and $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = 0$, then $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are in the same plane.

(4) Assume f is infinitely differentiable at 0 (i.e., $f^{(n)}(0)$ exists for all $n = 1, 2, 3, \ldots$). Then there exists $\varepsilon > 0$ such that

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

for all $-\varepsilon < x < \varepsilon$.

(5) Let $A(x) = \sum_n a_n x^n$ and $B(x) = \sum_n b_n x^n$ be power series with radii of convergence R_1 and R_2, respectively. Then $A(x)B(x)$ is again a power series with radius of convergence $\min\{R_1, R_2\}$.

Problem 2.27. (2021 Fall KAIST Calculus I) Let $\{\mathbf{x}_n\}$ be a sequence of vectors, where

$$\mathbf{x}_n = (x_n^{(1)}, x_n^{(2)}, x_n^{(3)}).$$

Let $\mathbf{x}$ be a vector. Prove or disprove:

(a) $\|\mathbf{x}_n - \mathbf{x}\| \to 0$ if $\sum_{k=1}^{n} \|\mathbf{x}_k - \mathbf{x}\| \le 1$ for all n.

(b) $\|\mathbf{x}_n - \mathbf{x}\| \to 0$ if

$$\lim_{n\to\infty} \frac{\|\mathbf{x}_{n+1} - \mathbf{x}\|}{\|\mathbf{x}_n - \mathbf{x}\|} = \rho < 1.$$

Problem 2.28. (2021 Fall KAIST Calculus I) How many domains are separately enclosed by the following polar equations?

(1) $r = 4\cos(2\theta)$

(2) $r = \dfrac{4}{2 - \cos\theta}$

(3) $r = 2(1 + \cos\theta)$

Problem 2.29. (2021 Fall KAIST Calculus I) Consider the curve

$$r = \cos 3\theta, \qquad -\frac{\pi}{6} \le \theta \le \frac{\pi}{6}.$$

(a) Find the slopes of the curve at the origin.

(b) Find the length of the curve, using the fact that

$$\int_0^{\pi/3} \sqrt{1 + 8\sin^2 3\theta}\, \mathrm{d}\theta \approx 2.23.$$

(c) Find the area enclosed by the curve.

Problem 2.30. (2021 Fall KAIST Calculus I) Your eye is at $(4, 0, 0)$. You are looking at a triangular plate whose vertices are at $(1, 0, 1)$, $(1, 1, 0)$, and $(-2, 2, 2)$. The line segment from $(1, 0, 0)$ to $(0, 2, 2)$ passes through the plate. What portion of the line segment is hidden from your view by the plate?

Problem 2.31. (2021 Fall KAIST Calculus I) Let

$$L_1: \ x = 1 + t,\ y = 3 - t,\ z = 2t, \qquad L_2: \ x = 0,\ y = -2s,\ z = 5s.$$

Find the minimum of $\|\overrightarrow{P_1P_2}\|$ for $P_1 \in L_1$ and $P_2 \in L_2$.

Problem 2.32. (2022 Fall POSTECH Calculus II) Let ℓ be a line in $\mathbb{R}^4$ passing through $(0, 1, -1, 3)$ and parallel to $(1, 0, 3, -1)$. If $P = (6, 5, 4, 2)$ is a point not on ℓ, find the distance between ℓ and P.

Problem 2.33. (2022 Fall POSTECH Calculus II) Let $f(x, y) = x^3 + 3y$.

(a) Find an equation of the plane tangent to the graph of f at $(2, 4)$.

(b) Find the linear approximation of $f(x, y)$ near $(2, 4)$, and use it to estimate $f(2.01, 4.03)$.

Problem 2.34. (2022 Fall POSTECH Calculus II) The Laplace operator Δ is defined by

$$\Delta = \partial_{xx} + \partial_{yy}.$$

Compute

$$\Delta \ln\left(\sqrt{x^2 + y^2}\right), \qquad (x, y) \ne (0, 0).$$

Problem 2.35. (2022 Fall POSTECH Calculus II)

(a) Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be linear. Show that there exist numbers C_j, $j = 1, \dots, n$, such that

$$f(\mathbf{x}) = \sum_{j=1}^{n} C_j x_j, \qquad \mathbf{x} = (x_1, \dots, x_n).$$

(b) Show that

$$f(\mathbf{x}) = \sum_{j=1}^{n} C_j x_j$$

is differentiable on $\mathbb{R}^n$. Find ∇f.

Problem 2.36. (2022 Fall POSTECH Calculus II) Determine whether each of the following functions is continuous, and prove your answers.

(a)

$$f(x, y) = \begin{cases} \dfrac{xy}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

(b)

$$f(x, y) = \begin{cases} \dfrac{x^2 y}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

Hint. Use polar coordinates.

Problem 2.37. (2022 Fall POSTECH Calculus II) Let

$$f(x, y) = \frac{1}{1 - xy}.$$

Find the second-order Taylor approximation to f at $(0, 0)$.

Problem 2.38. (2022 Fall POSTECH Calculus II) A line L_1 in $\mathbb{R}^3$ is given by

$$x = t + 1, \qquad y = t + 2, \qquad z = t + 3,$$

and a line L_2 in $\mathbb{R}^3$ is given by

$$x = 1 - s, \qquad y = 1 + s, \qquad z = 2s.$$

Find a point P on L_1 and a point Q on L_2 such that the square of the distance between P and Q is a local minimum with respect to t and s.

Problem 2.39. (2022 Fall POSTECH Calculus II) Let

$$f(x, y) = x^2 + 3y^2, \qquad g(x, y) = x^4 + 2y^2.$$

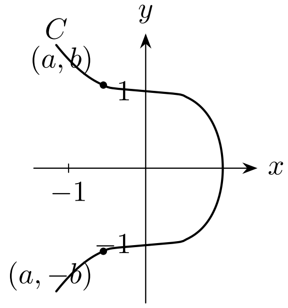
(a) Show that f and g attain their strict local minimum at the origin $(0, 0)$.

(b) Show that $\operatorname{Hess} f(0, 0)$ is positive definite but $\operatorname{Hess} g(0, 0)$ is not positive definite.

Problem 2.40. (2022 Fall POSTECH Calculus II) Let C be the curve in $\mathbb{R}^2$ defined by

$$x^3 + y^2 = 1.$$

The minimum distance from a point on C to the point $(-1, 0)$ occurs at $(a, \pm b) \in C$, where $b > 1$.



Check the conditions for using the Lagrange multiplier method. Find a using the Lagrange multiplier method.

Problem 2.41. (2022 Fall POSTECH Calculus II) Consider the function

$$f(x, y, z) = x^4 + y^4 + z^4 - 4xyz.$$

(a) Find all $P \in \mathbb{R}^3$ such that $\nabla f(P) = \mathbf{0}$.

(b) Show that at each local extremum except the origin, f has a local minimum.

(c) Show that $f(0, 0, 0)$ is not a local extremum.

Problem 2.42. (2022 Fall POSTECH Calculus II) Write a Riemann sum for

$$\iint_{[0,1]\times[0,1]} x^2 y^3 \, \mathrm{d}A$$

using points at the upper-right corner of each $\frac{1}{4}$-square.

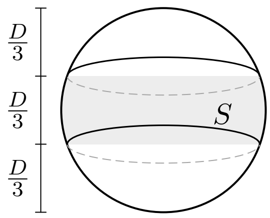
Problem 2.43. (2022 Fall POSTECH Calculus II) Let D be the parallelogram in $\mathbb{R}^2$ bounded by the lines

$$y = -x + 5, \qquad y = -x + 2, \qquad y = 2x - 1, \qquad y = 2x - 4.$$

Compute

$$\iint_D y^2 \, \mathrm{d}A.$$

Problem 2.44. (2022 Fall POSTECH Calculus II) An orange is cut into three slices with equal width.



Assume that the surface of the orange is a sphere with diameter D. Find the area of the peel from the middle slice (the area of S in the figure) by evaluating a surface integral explicitly.

Problem 2.45. (2022 Fall POSTECH Calculus II) Find

$$\oint_C \mathbf{F} \cdot \mathbf{T} \, \mathrm{d}s,$$

where

$$\mathbf{F}(x, y) = \left(-\frac{y}{x^2 + y^2}, \frac{x}{x^2 + y^2} \right),$$

and C is a loop that contains the origin in its interior and is traversed counterclockwise.

Problem 2.46. (2022 Fall POSTECH Calculus II) Let

$$\mathbf{F}(x, y, z) = (x^2 + y, \sin y, z).$$

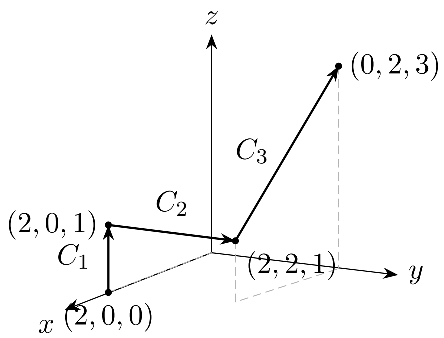
(a) Verify that $\mathbf{F}$ is not conservative.

(b) Verify that

$$\mathbf{G}(x, y, z) = (x^2, \sin y, z)$$

is conservative.

(c) Find the integral of $\mathbf{F}$ along the curve $C = C_1 \cup C_2 \cup C_3$ shown below, using part (b).



Problem 2.47. (2022 Fall POSTECH Calculus II) For each of the following statements, answer True or False. Here C is a smooth curve in $\mathbb{R}^2$, $-C$ is C traversed in the opposite direction, and $\mathbf{T}_C$ and $\mathbf{T}_{-C}$ are the unit tangent vectors to C and $-C$, respectively. The vector field $\mathbf{F}$ is continuously differentiable on $\mathbb{R}^2$.

(1) $\displaystyle\int_C f \, \mathrm{d}s = -\int_{-C} f \, \mathrm{d}s$, where $f \colon \mathbb{R}^2 \to \mathbb{R}$ is continuous.

(2) $\displaystyle\int_C \mathbf{F} \cdot \mathbf{T}_C \, \mathrm{d}s = -\int_{-C} \mathbf{F} \cdot \mathbf{T}_{-C} \, \mathrm{d}s.$

(3) $\displaystyle\int_C \mathbf{F}\cdot\mathbf{N}\ \mathrm{d}s = -\int_{-C} \mathbf{F}\cdot\mathbf{N}\ \mathrm{d}s$ with the same unit normal vector $\mathbf{N}$.

(4) $\mathbf{F}$ is conservative if and only if $\operatorname{curl}\mathbf{F} = 0$.

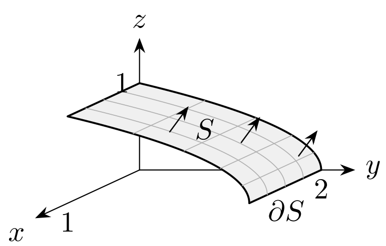
(5)
$$\int_{h(a)}^{h(b)} g(x)\ \mathrm{d}x = \int_a^b g(h(t))|h'(t)|\ \mathrm{d}t,$$
where $g\colon \mathbb{R}\to\mathbb{R}$ is continuous and $h\colon [a,b]\to\mathbb{R}$ is C^1.

Problem 2.48. (2022 Fall POSTECH Calculus II)

(a) State Stokes' theorem, including all conditions.

(b) Let
$$D = \left\{(x,y,z)\in\mathbb{R}^3 : x^2+y^2+z^2 = 5^2,\ y\le -1\right\}.$$
Find the area of D. Verify Stokes' theorem for the vector field
$$\mathbf{F}(x,y,z) = (-z,-4,x).$$

Problem 2.49. (2022 Fall POSTECH Calculus II) Let $\mathbf{F}(x,y,z) = (x,y,z)$, and let $S\subseteq\mathbb{R}^3$ be the surface defined by
$$y+2z^2 = 2,\qquad 0\le x\le 1,\qquad 0\le z\le 1.$$



(a) Find the outward flux of $\mathbf{F}$ across S by evaluating a surface integral.

(b) Find the counterclockwise circulation of $\mathbf{F}$ around ∂S.

Problem 2.50. (2022 Fall POSTECH Calculus II) Let $\mathbf{F} = (xy, z, 2y)$ be a vector field in $\mathbb{R}^3$, and let C be a curve of intersection of the plane
$$x+z = 5$$
and the cylinder
$$x^2+y^2 = 9.$$
Orient C so that its projection onto the xy-plane is oriented clockwise. Find
$$\int_C \mathbf{F}\cdot\mathbf{T}\ \mathrm{d}s.$$

Problem 2.51. (2022 Fall POSTECH Calculus II) Use each of the following theorems to show that, for a regular set $D\subseteq\mathbb{R}^2$ with boundary oriented counterclockwise,
$$\operatorname{area}(D) = -\int_{\partial D} y\ \mathrm{d}x.$$

(a) Use the Divergence theorem.

(b) Use Green's theorem.

Problem 2.52. (2022 Fall POSTECH Calculus II) Let

$$\mathbf{F}(x, y, z) = (y + z, z + x, x + y)$$

be a vector field in $\mathbb{R}^3$, and let C be a piecewise smooth curve from the origin to any point on the unit sphere

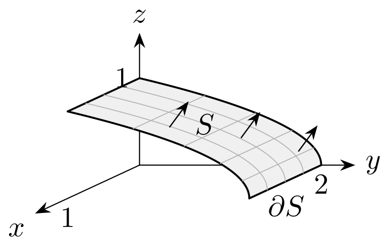
$$x^2 + y^2 + z^2 = 1.$$

Find the maximum of

$$\int_C \mathbf{F} \cdot \mathbf{T} \, \mathrm{d}s.$$

Problem 2.53. (2022 Fall POSTECH Calculus II) Let $\mathbf{F}(x, y, z) = (x, y, z)$, and let $S \subseteq \mathbb{R}^3$ be the surface defined by

$$y + 2z^2 = 2, \qquad 0 \le x \le 1, \qquad 0 \le z \le 1.$$



Using the Divergence theorem, find the outward flux of $\mathbf{F}$ across S.
Hint. Consider a three-dimensional solid whose boundary contains S.

Problem 2.54. (2022 Fall POSTECH Calculus II)

(a) Show that

$$\nabla \cdot (v\nabla u) = v\Delta u + \nabla v \cdot \nabla u$$

for all twice differentiable functions u and v on a regular set $D \subset \mathbb{R}^3$.

(b) Derive the following equality for a twice differentiable function u on D:

$$\iiint_D \left(u\Delta u + \|\nabla u\|^2\right) \mathrm{d}V = \iint_{\partial D} u\nabla u \cdot \mathbf{N} \, \mathrm{d}S,$$

where $\mathbf{N}$ is the unit outward normal vector on the boundary ∂D.

(c) Let u be a twice differentiable function satisfying

$$\Delta u = u \quad \text{in } D, \qquad u = 0 \quad \text{on } \partial D.$$

Show that $u \equiv 0$.

Problem 2.55. (2023 Fall POSTECH Calculus II) Find an equation for the plane tangent to

$$x - 3y^2 + z^2 = 7$$

at $(1, 1, 3)$.

Problem 2.56. (2023 Fall POSTECH Calculus II) Let L be the line containing $(1, 2, 1, 1)$ and $(1, 0, 2, 2)$ in $\mathbb{R}^4$. Let P be the hyperplane defined by

$$x + y + z + w = 1$$

in $\mathbb{R}^4$.

(a) Do L and P intersect each other?

(b) If the answer is yes, find the intersection point. Otherwise, find the distance between L and P.

Problem 2.57. (2023 Fall POSTECH Calculus II) Let $f\colon \mathbb{R}^2 \to \mathbb{R}$ be defined by

$$f(x, y) = |x^3 + y^3|.$$

(a) Find $\nabla f(0, 0)$.

(b) Is f differentiable at $(0, 0)$? Why or why not? Justify your answer.

Problem 2.58. (2023 Fall POSTECH Calculus II) Let $u\colon \mathbb{R}^2 \to \mathbb{R}$ be a C^2 function, and define

$$v(r, \theta) := u(r\cos\theta, r\sin\theta).$$

Show that

$$u_{xx} + u_{yy} = v_{rr} + \frac{1}{r}v_r + \frac{1}{r^2}v_{\theta\theta}.$$

Problem 2.59. (2023 Fall POSTECH Calculus II) Suppose the electric potential at (x, y) is given by

$$\ln\left(\sqrt{x^2 + y^2}\right).$$

(a) Find the rate of change of the potential at $(3, 4)$ toward the origin.

(b) Find the rate of change of the potential at $(3, 4)$ in a direction perpendicular to the direction given above. You may choose either of the two possible directions.

Problem 2.60. (2023 Fall POSTECH Calculus II) Suppose there is a function $\mathbf{F}\colon \mathbb{R}^4 \to \mathbb{R}^2$ with component functions $\mathbf{F} = (f_1, f_2)$ defined by

$$\begin{aligned} f_1(x, y, z, w) &= e^{x+2y-z+w} - (2x^2 - y^2 + z^2 - 3w^2), \\ f_2(x, y, z, w) &= \ln(1 + x^2 + y^2 + z^2 + w^2) + (3x + 4y - 5z + w). \end{aligned}$$

Let $P = (0, 0, 0, 0) \in \mathbb{R}^4$.

(a) Show that there is a differentiable function $\mathbf{G}\colon \mathbb{R}^2 \to \mathbb{R}^2$ defined on a small open set U around $(0, 0)$ satisfying $\mathbf{G}(0, 0) = (0, 0)$ and

$$\mathbf{F}(P) = \mathbf{F}(x, y, \mathbf{G}(x, y))$$

for $(x, y) \in U$.

(b) Using the linear approximation of $\mathbf{G}$, find an approximate solution (z, w) of the nonlinear system

$$\begin{aligned} 1 &= e^{0.1+2(0.1)-z+w} - (2(0.1)^2 - (0.1)^2 + z^2 - 3w^2), \\ 0 &= \ln(1 + (0.1)^2 + (0.1)^2 + z^2 + w^2) + (3(0.1) + 4(0.1) - 5z + w). \end{aligned}$$

Problem 2.61. (2023 Fall POSTECH Calculus II) Find the shortest distance from $(0, 0, 3)$ to the surface

$$z = x^2 - y^2.$$

Problem 2.62. (2023 Fall POSTECH Calculus II) Find the local maximum or minimum value of the function

$$x^3 + y^3 + z^3 - 3xy - 3yz - 3zx$$

defined on the domain

$$\{(x, y, z) : x > 0,\ y > 0,\ z > 0\}.$$

Problem 2.63. (2023 Fall POSTECH Calculus II) Show that the function

$$f(x, y) = x + y$$

subject to the constraint

$$x^3 + x + y^3 + y = 1$$

has at most one local extremum.

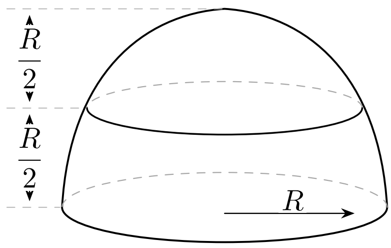
Problem 2.64. (2023 Fall POSTECH Calculus II) Let $D \subseteq \mathbb{R}^2$ be the domain given by

$$\frac{(x-1)^2}{4} + \frac{(y-3)^2}{9} \leq 1.$$

Find

$$\iint_D (x + y)\ \mathrm{d}A.$$

Problem 2.65. (2023 Fall POSTECH Calculus II) A solid hemisphere of radius R is split into two pieces with equal thickness. Use spherical coordinates to find the ratio of the volumes of the two pieces. Your answer should be a number between 0 and 1.



Problem 2.66. (2023 Fall POSTECH Calculus II) Let $\mathbf{F}$ be a vector field on $\mathbb{R}^2 \setminus \{(0, 0)\}$ defined by

$$\mathbf{F}(x, y) = \left(\frac{x}{x^2 + y^2}, \frac{y}{x^2 + y^2}\right).$$

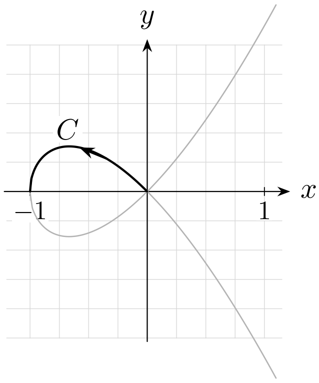
(a) Show that $\operatorname{curl} \mathbf{F} = 0$.

(b) Is $\mathbf{F}$ conservative? If so, find its potential. If not, explain why.

Problem 2.67. (2023 Fall POSTECH Calculus II) Let $C \subseteq \mathbb{R}^2$ be the curve defined by

$$y^2 = x^3 + x^2$$

in the second quadrant ($-1 \leq x \leq 0,\ y \geq 0$), oriented "to the left."



Find the work done by the force

$$\mathbf{F} = (2xy, -x^3)$$

along C.

Problem 2.68. (2023 Fall POSTECH Calculus II) Let S be the surface formed by the part of the paraboloid

$$z = 1 - x^2 - y^2$$

lying above the xy-plane, and let

$$\mathbf{F} = (x, y, 2(1 - z)).$$

Calculate the upward flux of $\mathbf{F}$ across S in each of the following ways:

(a) By direct calculation of

$$\iint_S \mathbf{F} \cdot \mathbf{N} \, \mathrm{d}S.$$

(b) Using the Divergence theorem and computing the flux across a simpler surface.

Problem 2.69. (2023 Fall POSTECH Calculus II) Let

$$\mathbf{F}(x, y, z) = (yz, -xz, 1).$$

Let S be the portion of the surface of the paraboloid

$$z = 4 - x^2 - y^2$$

which lies above the first octant, and let C be the closed curve

$$C = C_1 + C_2 + C_3,$$

where C_1, C_2, C_3 are the three curves formed by intersecting S with the xy-, yz-, and xz-planes, respectively, so that C is the boundary of S. Orient C so that it is traversed counterclockwise when seen from above in the first octant. Evaluate

$$\int_C \mathbf{F} \cdot \mathbf{T} \, \mathrm{d}s.$$

Problem 2.70. (2023 Fall POSTECH Calculus II) Let $\mathbf{F}$ be the vector field defined by

$$\mathbf{F}(x, y, z) = \left(\frac{z}{x^2 + z^2}, y, -\frac{x}{x^2 + z^2} \right).$$

Let C be the intersection of

$$x^2 + z^2 = 4 \qquad \text{and} \qquad x + y + z = 0$$

in $\mathbb{R}^3$. Find the circulation of $\mathbf{F}$ over C, oriented from $(2, -2, 0)$ to $(0, -2, 2)$, counterclockwise when viewed from the positive y-axis.

Problem 2.71. (2023 Fall POSTECH Calculus II) Two point charges are placed at

$$P_1 = (1, 0, 0) \qquad \text{and} \qquad P_2 = (-1, 0, 0),$$

with electric charges q_1 and q_2, respectively. The electric field $\mathbf{F}$ on $\mathbb{R}^3 \setminus \{P_1, P_2\}$ exerted by these two particles is given by

$$\mathbf{F}(x, y, z) = \frac{q_1}{4\pi\varepsilon_0} \frac{(x-1, y, z)}{\left((x-1)^2 + y^2 + z^2\right)^{3/2}} + \frac{q_2}{4\pi\varepsilon_0} \frac{(x+1, y, z)}{\left((x+1)^2 + y^2 + z^2\right)^{3/2}},$$

where ε_0 is the electric constant. If S is the ellipsoid

$$\frac{x^2}{2} + y^2 + z^2 = 1,$$

write the outward flux

$$\iint_S \mathbf{F} \cdot \mathbf{N} \, \mathrm{d}S$$

in terms of q_1 and q_2. Justify your answer.

Problem 2.72. (2024 Fall POSTECH Calculus II) Find parametric equations for the line of intersection of the planes

$$3x + 2y - z = 5 \qquad \text{and} \qquad x + y + z = 2.$$

Also find the distance from the point $S(1, 2, 2)$ to this line.

Problem 2.73. (2024 Fall POSTECH Calculus II) Find the directional derivative of

$$f(x, y, z) = xy^2 + yz^3$$

at $P(2, -1, 1)$ in the direction of $\overrightarrow{PQ}$, where $Q = (3, 1, 3)$.

Problem 2.74. (2024 Fall POSTECH Calculus II) Explicitly compute

$$\nabla \cdot (\nabla f) \qquad \text{and} \qquad \nabla \times (\nabla f)$$

for

$$f(x, y, z) = x^3 + y^3 + z^3 - 3xyz.$$

Problem 2.75. (2024 Fall POSTECH Calculus II) Let $D, E \subseteq \mathbb{R}^n$. Prove the following:

(a) If D and E are open, then $D \cap E$ is open.

(b) If $D \subseteq E$, then $\operatorname{int} D \subseteq \operatorname{int} E$, where $\operatorname{int} D$ denotes the interior of D.

Problem 2.76. (2024 Fall POSTECH Calculus II) Let $f \colon \mathbb{R} \to \mathbb{R}$ be continuous. Show that for any $i = 1, \dots, n$, the function

$$F_i \colon \mathbb{R}^n \to \mathbb{R}, \qquad (x_1, \dots, x_n) \mapsto f(x_i),$$

is continuous.

Problem 2.77. (2024 Fall POSTECH Calculus II)

(a) Show that

$$g(x,y) = \begin{cases} \dfrac{x^2 - y^2}{x^2 + y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0), \end{cases}$$

is not continuous at $(0,0)$.

(b) Show that

$$f(x,y) = \begin{cases} \dfrac{xy(x^2 - y^2)}{x^2 + y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0), \end{cases}$$

is differentiable at $(0,0)$.

Problem 2.78. (2024 Fall POSTECH Calculus II) Let $x, y \colon \mathbb{R}^2 \to \mathbb{R}$ be functions defined by

$$x(u,v) = u^3 - 3uv^2,$$
$$y(u,v) = 3u^2 v - v^3.$$

Also define

$$f(x,y) = e^{x^2 - y^2} \cos(2xy).$$

(a) Show that

$$x_u = y_v, \qquad x_v = -y_u, \qquad f_{xx} + f_{yy} = 0.$$

(b) Let

$$g(u,v) := f(x(u,v), y(u,v)).$$

Compute $g_{uu} + g_{vv}$.

Problem 2.79. (2024 Fall POSTECH Calculus II) Find all the local maxima, local minima, and saddle points of

$$f(x,y) = e^x (2x^2 - y^2).$$

Problem 2.80. (2024 Fall POSTECH Calculus II) Let

$$T(x,y,z) = 2x^2 + 4yz - 8z + 300$$

be the temperature at the point (x,y,z). Find the hottest and coldest points on the ellipsoid

$$x^2 + y^2 + 2z^2 = 4$$

and their temperatures. Use the Lagrange multiplier method.

Problem 2.81. (2024 Fall POSTECH Calculus II) Find the order 2 Taylor approximation of

$$f(x, y, z) = \cos(x + y + z)$$

at $P = (0, 0, 0)$, and use it to approximate

$$f(0.1, 0.1, 0.1) = \cos(0.3).$$

Also find an error bound for the approximation.

Problem 2.82. (2024 Fall POSTECH Calculus II) Let $\mathbf{F}\colon \mathbb{R}^3 \to \mathbb{R}^2$, $\mathbf{F} = (f_1, f_2)$, be defined by

$$f_1(x, y, z) = x^2 + y^2 - z^2,$$
$$f_2(x, y, z) = \frac{x^2}{5} + \frac{y^2}{2} + \frac{z^2}{2}.$$

Let

$$P = \left(1, \frac{2}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right).$$

Then the level set

$$C = \{(x, y, z) \in \mathbb{R}^3 \mid \mathbf{F}(x, y, z) = \mathbf{F}(P)\}$$

defines a curve in $\mathbb{R}^3$.

(a) Show that there is a continuously differentiable function $\mathbf{G}\colon \mathbb{R} \to \mathbb{R}^2$, $\mathbf{G} = (g_1, g_2)$, such that every point in C sufficiently close to P can be expressed as

$$(x, y, z) = (\mathbf{G}(z), z).$$

(b) Find any unit tangent vector tangent to C at P.

3 (MATH311) Analysis I

4 (MATH312) Analysis II